

A resolution-free proof of the Duistermaat–van der Kallen nonvanishing theorem

Christopher D. Long*

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Abstract

We give a resolution-free semialgebraic proof of the Duistermaat–van der Kallen theorem: if the Newton polytope of a nonzero complex Laurent polynomial contains the origin, infinitely many positive powers have nonzero constant term. A uniform middle-dimensional sublevel estimate for a restricted holomorphic volume form yields logarithmic-form integrability by dyadic decomposition. For global transport, we use the closed embedding $z \mapsto (z, z^{-1})$ and its induced Hermitian metric. Uniform semialgebraic connecting-path bounds imply finiteness of scalar asymptotic critical values; an explicit normalized-gradient lift then has complete transport by a linear-growth estimate. The local residue construction of Duistermaat and van der Kallen identifies the constant-term generating function with a fiber period. A separate semialgebraic endpoint sweep gives a primitive with a power-law bound, contradicting the simple pole forced by vanishing of all positive constant terms. Infinite nonvanishing follows by applying minimal nonvanishing to powers. The standard semialgebraic, integration, and ODE inputs are identified; an independent Whitney–Thom transport proof is retained in an appendix.

1 Introduction and statements

Let $d \geq 0$ and

$$R_d = \mathbb{C}[x_1^{\pm 1}, \dots, x_d^{\pm 1}].$$

For $0 \neq f = \sum_{\alpha \in \mathbb{Z}^d} a_\alpha x^\alpha \in R_d$, define

$$A = \text{supp}(f) = \{\alpha : a_\alpha \neq 0\}, \quad \text{Newt}(f) = \text{conv}(A), \quad c_n(f) = \text{CT}(f^n) \quad (n \geq 0).$$

In particular $c_0(f) = 1$. For $d = 0$, the ring is $\mathbb{C}$ and the exponent space is the zero-dimensional vector space.

Theorem 1.1 (Duistermaat–van der Kallen, nonvanishing). *Let $d \geq 0$ and $0 \neq f \in R_d$. If $0 \in \text{Newt}(f)$, then for every integer $M \geq 0$ there is an integer $n > M$ such that $c_n(f) \neq 0$.*

Theorem 1.2 (Minimal nonvanishing). *Let $d \geq 0$ and $0 \neq f \in R_d$. If $0 \in \text{Newt}(f)$, there exists $n \geq 1$ such that $c_n(f) \neq 0$.*

*Email: galizur@gmail.com.

Theorem 1.1 follows algebraically from Theorem 1.2, by applying it to f^{M+1} ; see Section 9. Thus only the minimal statement needs an analytic proof. Strict separation subsequently gives

$$(\text{CT}(f^n) = 0 \text{ for every } n \geq 1) \implies \text{CT}(hf^n) = 0 \text{ for all sufficiently large } n$$

for every fixed Laurent polynomial h . The corresponding assertion for finite Fourier sums on a compact torus follows by integrating characters.

The original proof is due to Duistermaat and van der Kallen [3]. Kiselman's Question 9.1, recording a question of Passare, asks for a simpler proof of infinite nonvanishing [9]. The argument here is organized around a direct estimate for a restricted complex differential form. Its local residue construction and primitive-differentiation identity come from [3, Sections 4–5]; the new argument to be examined is the sublevel estimate and its use to control chains that may escape to infinity. We make no claim of bibliographic priority for individual auxiliary lemmas.

Proof architecture and scope

After the elementary face reduction, assume that $d \geq 1$ and $0 \in \text{int Newt}(f)$. Put

$$\Omega_d = dz_1 \wedge \cdots \wedge dz_d, \quad \omega_d = \frac{dx_1}{x_1} \wedge \cdots \wedge \frac{dx_d}{x_d}.$$

The argument has the following implications:

uniform finite-multiplicity projection estimate

$$\begin{aligned} \implies & \int_{E \cap \{|P| \leq \varepsilon\}} |\Omega_d| \leq C\varepsilon^{1/D} \\ \implies & \int_{S \cap \{|f| \leq R\}} |\omega_d| \leq C_{S,f} R^{1/D} \\ \implies & \left| \int_a^{2a} J(s) ds \right| \longrightarrow 0 \quad (a \downarrow 0). \end{aligned}$$

If all positive constant terms vanish, the residue identity forces $J(s) = -(2\pi i)^{d-1}/s$. Its integral over $[a, 2a]$ is a fixed nonzero constant, giving a contradiction.

Global transport is established separately by the implications

Hardt triviality, compact triangulation, and uniform arc lengths

$$\begin{aligned} \implies & \text{finitely many scalar asymptotic critical values} \\ \implies & \text{a normalized-gradient lift with controlled growth} \\ \implies & \text{complete fiber transport outside a finite set.} \end{aligned}$$

The radius and differential norm here come from the closed embedding $z \mapsto (z, z^{-1})$, so approaching a deleted coordinate hyperplane counts as escape to infinity. The resulting smooth flow transports homology classes; it is not asserted to be semialgebraic. Hardt triviality is used both in the connecting-path lemma and, later, to choose the semialgebraic sweep on the final real interval.

The term *resolution-free* is literal: no resolution of singularities is used. The main proof retains semialgebraic decomposition, Hardt triviality, compact triangulation, integration and Stokes, together with ordinary finite-dimensional ODE existence, dependence, and continuation. It does not invoke Whitney stratification, Thom isotopy, or a generalized Bertini–Sard theorem. Appendix A gives an independent Whitney–Thom proof of the finite-exceptional-set and topological transport conclusion needed for the period argument. The presentation works relative to the explicitly stated standard inputs; it is not a reconstruction of their foundations. The stronger critical-value and exponential-growth statement in [3, Theorem 5] is not a theorem of this draft.

2 Elementary Newton-polytope reductions

Lemma 2.1 (Minimal-face reduction). *Suppose $0 \in \text{Newt}(f)$. There is a Laurent polynomial $\tilde{f}$ in $r \leq d$ variables such that*

$$\text{CT}(\tilde{f}^n) = \text{CT}(f^n) \quad (n \geq 0),$$

and either $r = 0$ and $\tilde{f}$ is a nonzero constant, or $r \geq 1$ and $0 \in \text{int Newt}(\tilde{f})$ in $\mathbb{R}^r$.

Proof. Let Q be the smallest face of $\text{Newt}(f)$ containing 0 . Then $0 \in \text{relint } Q$. If Q is proper, choose an exposing linear functional ℓ with $\ell = 0$ on Q and $\ell > 0$ on $\text{Newt}(f) \setminus Q$. This is possible because the minimum occurs at $0 \in Q$. If Q is the whole polytope, use $\ell = 0$.

Let f_Q consist of the terms of f with exponent in Q . A sum of n support exponents can equal 0 only if every summand belongs to Q : apply ℓ and use nonnegativity. Comparing the full coefficient expansions, including all coefficients and cancellations, gives $\text{CT}(f^n) = \text{CT}(f_Q^n)$.

The affine span L of Q is linear because $0 \in Q$. Since Q is a lattice polytope, L is rational and $\Lambda = L \cap \mathbb{Z}^d$ is a lattice of rank $r = \dim L$. Choose a $\mathbb{Z}$ -basis of Λ and express f_Q in the corresponding group algebra $\mathbb{C}[\Lambda] \simeq R_r$. The zero exponent and all products are preserved. If $r = 0$, its sole coefficient is nonzero. Otherwise the image of Q has 0 in its full-dimensional interior. $\square$

Lemma 2.2 (Powers of Newton polytopes). *For $0 \neq f \in R_d$ and $q \geq 1$,*

$$\text{Newt}(f^q) = q \text{Newt}(f).$$

Proof. Every exponent of f^q is a sum of q exponents of f , giving one inclusion. Let v be a vertex of $\text{Newt}(f)$, and choose a linear functional uniquely maximized at v on the support. If $\alpha_1 + \dots + \alpha_q = qv$, equality at this maximum forces $\alpha_1 = \dots = \alpha_q = v$. Thus the coefficient at qv is $a_v^q \neq 0$. All vertices of $q \text{Newt}(f)$ occur in f^q , proving the opposite inclusion. The singleton case is immediate. $\square$

Lemma 2.3 (A unimodular chart at a vertex). *Suppose $d \geq 1$ and $0 \in \text{int Newt}(f)$. After a unimodular monomial change of variables, which preserves every constant term of every power, one can write*

$$f(y) = y_1^{-m_1} \dots y_d^{-m_d} u(y), \quad m_i \in \mathbb{Z}_{>0}, \quad u \in \mathbb{C}[y_1, \dots, y_d], \quad u(0) \neq 0. \quad (2.1)$$

Proof. Choose a vertex v of $\text{Newt}(f)$ and let C be the open cone of real linear functionals whose minimum on the support occurs uniquely at v . Choose a primitive lattice vector $a_1 \in C$, and extend it to a lattice basis $a_1, b_2, \dots, b_d$ of the dual exponent lattice. Such a primitive vector exists by rational density in the nonempty open cone, followed by division by the gcd of its coordinates.

For sufficiently large integers N_i , the vectors $a_i = b_i + N_i a_1$ lie in C : divide by N_i and use openness. These vectors together with a_1 still form a lattice basis. In the resulting exponent coordinates every support exponent α satisfies $\langle a_i, \alpha \rangle \geq \langle a_i, v \rangle$, with simultaneous equality only at v . Since 0 is an interior point, every nonzero linear functional takes negative values on the support; hence $\langle a_i, v \rangle < 0$. Set $m_i = -\langle a_i, v \rangle$. Factoring out y^{-m} gives (2.1) with constant coefficient a_v .

The exponent change is a lattice automorphism, so it preserves zero exponents, products, and constant terms. From this point one may regard f as the transformed Laurent polynomial and use the standard ordered form in the y -coordinates; no unrecorded orientation sign is then introduced. $\square$

3 Semialgebraic integration and uniform projections

Semialgebraic means semialgebraic over $\mathbb{R}$, with arbitrary real parameters allowed. Complex coordinates are identified with their real and imaginary parts. Dimension in this section is real dimension.

3.1 The standard inputs

The following precise forms of standard results will be used.

Standard input 3.1 (Semialgebraic decomposition and triviality). Coordinate projections of semialgebraic sets are semialgebraic. A semialgebraic subset of the real line is a finite union of points and intervals; in particular, if it is unbounded above it contains a tail. A semialgebraic set and a finite collection of semialgebraic subsets admit compatible finite decompositions into Nash manifolds. Semialgebraic families have uniform bounds on the number of connected components of their fibers. For a continuous semialgebraic map $g : X \rightarrow Y$, there is a finite semialgebraic partition $Y = \coprod_j Y_j$ and semialgebraic homeomorphisms

$$g^{-1}(Y_j) \simeq Y_j \times F_j$$

commuting with projection to Y_j . Compatible triangulations may be chosen for compact semialgebraic sets. Compact singular homology classes on semialgebraic manifolds admit finite semialgebraic cycle representatives.

These are the decomposition, triangulation, and Hardt triviality results in [1, 7]; the local-triviality statement is [1, Theorem 9.3.2]. A noncompact set need not have a finite triangulation by *closed* simplices. Only compact model fibers, compact cycle representatives, and compact truncated chains will require that assertion here.

Standard input 3.2 (Integration and Stokes). Integration of a smooth form over a finite semialgebraic chain is compatible with finite subdivision and semialgebraic reparametrization. A bounded semialgebraic set of dimension k has finite k -dimensional Hausdorff measure. For a compact semialgebraic chain C and a smooth $(k-1)$ -form η on a neighborhood of its support,

$$\int_C d\eta = \int_{\partial C} \eta.$$

On a smooth semialgebraic manifold, integration of a closed form over semialgebraic cycles agrees with the usual homology pairing.

The integration convention and change of variables are described in [6, Sections 1–2]; Stokes is [6, Theorem 2.9], extended to finite chains by subdivision and cancellation of common faces. Compatible C^1 triangulations give another foundation [11]. The finite-volume assertion can also be obtained by summing the coordinate-projection bounds proved below. We use ordinary change of variables on smooth pieces, not a slicing theorem for currents.

3.2 The absolute density of a complex form

Definition 3.3. Let $S \subset \mathbb{R}^N$ be semialgebraic of dimension at most k , and let ξ be a smooth complex-valued k -form on a neighborhood of S . On a smooth k -dimensional stratum M , let $\tau_M(x)$ be either unit simple k -vector spanning $T_x M$. Define

$$\int_S |\xi| = \sum_{\dim M=k} \int_M |\xi_x(\tau_M(x))| d\mathcal{H}^k(x). \quad (3.1)$$

Lower-dimensional strata contribute zero. For $k=0$ this is counting measure. For a finite chain $C = \sum_j n_j C_j$, the positive quantity used for estimates is $\sum_j |n_j| \int_{C_j} |\xi|$; we write it as $\int_{|C|} |\xi|$. The oriented integral is linear in the chain.

The definition is independent of the chosen smooth decomposition, by common refinement and the change-of-variables formula. It satisfies

$$|g\xi| = |g| |\xi|, \quad |\xi + \zeta| \leq |\xi| + |\zeta|, \quad \left| \int_C \xi \right| \leq \int_{|C|} |\xi|. \quad (3.2)$$

The first identity concerns the modulus of the *full* restricted complex form. Expanding into real coordinate wedges is permissible for upper bounds, not as an alternative definition.

For example, on $E = \{(z, -z) : 1 \leq |z| \leq 2\}$ one has $(dz_1 \wedge dz_2)|_E = 0$, although the sum of the absolute values of its separate real coordinate terms is nonzero. This cancellation will be essential in Section 4.

3.3 The uniform projection estimate

Lemma 3.4 (Uniform finite-multiplicity integration). *Let Θ be semialgebraic and let $\mathcal{E} \subset \Theta \times \mathbb{R}^N$ be a semialgebraic family with $\dim E_\theta \leq k$. Suppose $h_\theta : E_\theta \rightarrow \mathbb{R}^k$ is a semialgebraic family of continuous maps, differentiable on the smooth pieces after decomposition. There is $N_0 < \infty$, depending on this entire family, such that*

$$\int_{E_\theta} |dh_{\theta,1} \wedge \cdots \wedge dh_{\theta,k}| \leq N_0 \operatorname{vol}_k(h_\theta(E_\theta)) \quad (\theta \in \Theta). \quad (3.3)$$

An unbounded image is allowed; the right side may be infinite.

Proof. Retain the parameter in the map

$$H : \mathcal{E} \longrightarrow \Theta \times \mathbb{R}^k, \quad H(\theta, x) = (\theta, h_\theta(x)).$$

By Standard input 3.1, the numbers of connected components of all its fibers are bounded by one integer N_0 . In particular the cardinality of every zero-dimensional fiber is at most N_0 .

Fix θ . Decompose E_θ into smooth pieces on which the rank of h_θ is constant. Pieces of dimension less than k , and pieces on which the rank is less than k , have zero pullback k -density. On rank- k pieces of dimension k , the map is locally a diffeomorphism. The set of target points whose whole fiber has positive dimension has dimension at most $k - 1$; its rank- k inverse images also have measure zero, by local change of variables. Away from this exceptional set there are at most N_0 preimages in total. Apply ordinary change of variables on injective smooth pieces, or equivalently the area formula with this counting multiplicity, and sum to obtain (3.3). This is the parameter-uniform version of [6, Proposition 2.5]; the parameter-retaining device also appears in the proof of its Proposition 2.8. $\square$

Remark 3.5. In applications, θ includes all varying set parameters, polynomial coefficients, and cutoffs. Parameters need not range in a compact set. Uniformity follows from one fixed semialgebraic description, not from compactness of parameter space. Adding finitely many prescribed semialgebraic inequalities produces another fixed family, to which the same lemma applies with a possibly different constant.

Corollary 3.6. *For a semialgebraic family E_θ of dimension at most d contained in a fixed bounded box in $\mathbb{C}^d$,*

$$\sup_\theta \int_{E_\theta} |\Omega_d| < \infty.$$

More generally, the Euclidean d -volumes of a bounded semialgebraic family of subsets of $\mathbb{R}^N$ of dimension at most d are uniformly bounded.

Proof. Expand Ω_d into its 2^d real coordinate wedge terms and use (3.2) and Lemma 3.4 for the corresponding coordinate projections. Their images lie in fixed boxes. For the Euclidean-volume assertion, the coordinate components of a unit simple d -vector have squares summing to 1, so their absolute values have sum at least 1. Sum over all coordinate d -projections. $\square$

3.4 Uniform lengths of connecting paths

Lemma 3.7 (Uniform connecting paths). *Let $\mathcal{A} \subset \Theta \times \mathbb{R}^m$ be a fixed semialgebraic family whose fibers A_θ are compact and contained in the closed unit ball. There are integers $N \geq 1$ and a constant $L \geq 1$, depending only on the family, such that each fiber has at most N connected components and any two points in the same component can be joined in that component by a rectifiable semialgebraic path of length at most L .*

Proof. The component bound is part of Standard input 3.1. Apply Hardt triviality to the parameter projection. After a finite partition of Θ , each nonempty family has a semialgebraic trivialization with one fixed compact model fiber. Triangulate that model by a finite simplicial complex K , and compose its triangulation with the trivialization to obtain semialgebraic homeomorphisms $H_\theta : |K| \rightarrow A_\theta$, jointly semialgebraic in θ and the model point.

For every simplex σ of K , fix a vertex v_σ . Given two points p, q of the same component of $|K|$, choose simplices containing them. Join p to the chosen vertex of its simplex by a straight segment, join the two chosen vertices through the one-skeleton of their connected component, and join the second vertex to q . A connected finite complex has a connected one-skeleton on its vertices. There are finitely many pairs of vertices, so these edge paths may be fixed in advance with a uniform bound on the number of segments. The two endpoint segments add at most two more.

Under H_θ , each segment is constant or an injectively parametrized semialgebraic arc in the unit ball. As θ and the endpoint in a fixed simplex vary, its image belongs to one semialgebraic family of sets of dimension at most one. Indeed it is the image of $\{(\theta, p, t) : p \in \sigma, 0 \leq t \leq 1\}$ under $(\theta, p, t) \mapsto (\theta, p, H_\theta((1-t)p + tv_\sigma))$. The fixed edge types give finitely many more such families. Corollary 3.6, in real dimension one, bounds the one-dimensional Hausdorff measure of all these images uniformly. The length of an injective arc is the measure of its image. Summing these bounds over the bounded number of segments, and then taking the maximum over the finitely many Hardt pieces, gives L .

The transported parametrizations need not have bounded derivatives. Rectifiability is what the argument establishes and uses: each arc can be parametrized by arclength, and decomposed into smooth pieces for the ordinary chain rule. No Lipschitz bound for the Hardt homeomorphisms is assumed. $\square$

Remark 3.8. A more general bounded-family path-length theorem is stated as [13, Theorem 3.3], attributed there to Teissier. The proof above supplies the compact-fiber case needed here directly from the semialgebraic tools and projection bound already used in this paper. It does not invoke a resolution-based proof of that theorem.

4 A middle-dimensional polynomial sublevel estimate

Fix $d \geq 1$, $L \geq 1$, a semialgebraic family E_θ of real dimension at most d in the box $[-L, L]^{2d} \subset \mathbb{C}^d$, and $D \geq 1$. For a polynomial P , write

$$\|P\|_{\text{coef}} = \max_{\alpha} |[z^\alpha]P|.$$

Theorem 4.1 (Uniform sublevel estimate). *There is $C = C(\mathcal{E}, d, D, L)$ such that every holomorphic polynomial $P \in \mathbb{C}[z_1, \dots, z_d]$ with $\deg P \leq D$ and $\|P\|_{\text{coef}} = 1$ satisfies*

$$\int_{E_\theta \cap \{|P| \leq \varepsilon\}} |\Omega_d| \leq C\varepsilon^{1/D} \quad (\theta \in \Theta, \varepsilon > 0). \quad (4.1)$$

The same estimate, with value zero on the left, holds for $\varepsilon = 0$.

Lemma 4.2 (The one-derivative estimate). *For each $i \in \{1, \dots, d\}$, let*

$$\Xi_i(P) = dP \wedge dz_1 \wedge \cdots \widehat{dz_i} \cdots \wedge dz_d.$$

There is C_1 , uniform in $\theta, P, \varepsilon, r$ and i , such that

$$\int_{E_\theta \cap \{|P| \leq \varepsilon, |\partial_i P| \geq r\}} |\Xi_i(P)| \leq C_1 \varepsilon \quad (r > 0, \varepsilon > 0). \quad (4.2)$$

The assertion also holds when the derivative cutoff is omitted.

Proof. Expand $\Xi_i(P)$ into real wedges. Each term is the pullback of real coordinate volume under a map whose first coordinate is $\Re P$ or $\Im P$, and whose other coordinates are one real coordinate from each z_j with $j \neq i$. The image of the indicated set lies in

$$[-\varepsilon, \varepsilon] \times [-L, L]^{d-1}.$$

Apply Lemma 3.4 to the family retaining θ , the real and imaginary coefficients of P , ε , and r . There are only finitely many choices of i and real wedge term. Their image volumes are at most $2\varepsilon(2L)^{d-1}$. Summing proves the assertion. The triangle inequality is applied only after evaluating each complex form on the tangent vector. $\square$

Proof of Theorem 4.1. We induct on the degree bound D . Corollary 3.6 gives a bound V_0 for the integral over all of E_θ . It suffices to handle $0 < \varepsilon < 1/2$; enlarging the final constant handles $\varepsilon \geq 1/2$.

Let $H = \max\{1, \sqrt{2}L\}$ and

$$B_D = \sum_{0 < |\alpha|_1 \leq D} H^{|\alpha|_1}, \quad c = (2B_D)^{-1}.$$

If every nonconstant coefficient of P has modulus less than c , its constant coefficient has modulus 1 and the remaining terms have sum of moduli at most $1/2$ throughout the box. Thus $|P| \geq 1/2$ there, and the sublevel set under consideration is empty.

Otherwise choose a nonconstant coefficient $[z^\beta]P$ of modulus at least c , and an i with $\beta_i > 0$. Put $Q = \partial_i P$ and $M = \|Q\|_{\text{coef}}$. Distinct monomials remain distinct under this partial differentiation whenever they survive; hence

$$M \geq \beta_i |[z^\beta]P| \geq c. \quad (4.3)$$

Holomorphy gives the exact identity

$$\Xi_i(P) = (-1)^{i-1} Q \Omega_d. \quad (4.4)$$

By (3.2), on $|Q| \geq r$,

$$|\Omega_d| \leq r^{-1} |\Xi_i(P)|.$$

Consequently Lemma 4.2 bounds the large-derivative part by $C_1 \varepsilon / r$.

If $D = 1$, then Q is a constant of modulus at least c . Applying the same argument with $r = c$ gives the bound $C_1\varepsilon/c$. This establishes the base case, including polynomials of degree zero through the preceding near-constant case.

Suppose $D \geq 2$. The induction hypothesis applies to the normalized polynomial Q/M of degree at most $D - 1$, even when it is constant. Discarding the condition $|P| \leq \varepsilon$ on the small-derivative part, we obtain

$$\begin{aligned} \int_{E_\theta \cap \{|P| \leq \varepsilon\}} |\Omega_d| &\leq C_1 \frac{\varepsilon}{r} + C_{D-1} \left(\frac{r}{M} \right)^{1/(D-1)} \\ &\leq C_1 \frac{\varepsilon}{r} + C_{D-1} c^{-1/(D-1)} r^{1/(D-1)}. \end{aligned} \quad (4.5)$$

Take $r = \varepsilon^{(D-1)/D}$. Both terms are constant multiples of $\varepsilon^{1/D}$, proving the induction. All constants depend only on the fixed family and degree bounds. Finally, let $\varepsilon \downarrow 0$ and use monotonicity of the positive integral to see that the integral over $P = 0$ is zero. $\square$

Remark 4.3. This is a sublevel estimate for a restricted complex top form, not for Euclidean volume of E_θ . The distinction permits the estimate when a whole component of E_θ lies in $P = 0$: on that component the relevant complex form must have zero integral density. No regularity or transversality of the hypersurface $P = 0$ is assumed.

5 Logarithmic integrability on the torus

Theorem 5.1 (Middle-dimensional logarithmic integrability). *Let $d \geq 1$, let $f \in R_d$ satisfy $0 \in \text{int Newt}(f)$, and let $S \subset (\mathbb{C}^\times)^d$ be semialgebraic of real dimension at most d . Choose $b \in \mathbb{Z}_{\geq 0}^d$ with $b + \text{supp}(f) \subset \mathbb{Z}_{\geq 0}^d$ and put*

$$D = \max_{\alpha \in \text{supp}(f)} |b + \alpha|_1.$$

Then $D \geq 1$, and there is a constant $C = C(S, f, b)$ such that

$$\int_{S \cap \{|f| \leq R\}} |\omega_d| \leq CR^{1/D} \quad (R > 0). \quad (5.1)$$

The same assertion holds for a finite semialgebraic chain, with absolute multiplicities. In particular, the integral of $|\omega_d|$ is finite on every such chain on which f is bounded.

Proof. Interiority gives a $\kappa > 0$ such that $[-\kappa, \kappa]^d \subset \text{Newt}(f)$. With $a_* = \min_{\alpha \in \text{supp}(f)} |a_\alpha| > 0$, define

$$\rho_k = \max_{\alpha \in \text{supp}(f)} |a_\alpha| 2^{\langle k, \alpha \rangle}, \quad k \in \mathbb{Z}^d.$$

The support function of the contained cube gives

$$\rho_k \geq a_* 2^{\kappa \|k\|_1}. \quad (5.2)$$

Partition the torus into the disjoint half-open boxes

$$B_k = \{x : 2^{k_i} \leq |x_i| < 2^{k_i+1} \ (1 \leq i \leq d)\}.$$

On B_k write $x_i = 2^{k_i} z_i$, so $1 \leq |z_i| < 2$. Positive diagonal scaling preserves the logarithmic form. The polynomial

$$P_k(z) = \rho_k^{-1} z^b f(2^{k_1} z_1, \dots, 2^{k_d} z_d)$$

has degree at most D and coefficient norm exactly 1. There is no combination of distinct exponents in this normalization. If $|f| \leq R$, then

$$|P_k(z)| \leq 2^{|b|_1} R / \rho_k. \quad (5.3)$$

The rescaled sets belong to one semialgebraic family

$$E_\lambda = \{z : 1 \leq |z_i| < 2 \text{ for every } i, (\lambda_1 z_1, \dots, \lambda_d z_d) \in S\}, \quad \lambda \in (0, \infty)^d.$$

This family lies in a fixed bounded box and has dimension at most d . It includes every set obtained from $S \cap B_k$. On its fixed annulus,

$$\omega_d = (z_1 \cdots z_d)^{-1} \Omega_d, \quad |\omega_d| \leq |\Omega_d|.$$

Applying Theorem 4.1 with this entire family, and then (5.2), gives a constant C_0 independent of k and R with

$$\begin{aligned} \int_{S \cap B_k \cap \{|f| \leq R\}} |\omega_d| &\leq C_0 (2^{|b|_1} R / \rho_k)^{1/D} \\ &\leq C_1 R^{1/D} 2^{-\kappa \|k\|_1 / D}. \end{aligned} \quad (5.4)$$

Here $D = 0$ would force the support of f to be a singleton, contradicting full-dimensional interiority for $d \geq 1$.

Set $q = 2^{-\kappa/D} \in (0, 1)$. The convergent sum is explicitly

$$\sum_{k \in \mathbb{Z}^d} q^{\|k\|_1} = \left(1 + 2 \sum_{j \geq 1} q^j\right)^d = \left(\frac{1+q}{1-q}\right)^d.$$

Countable additivity of the nonnegative integral on the disjoint boxes proves (5.1). Boundary pieces are assigned to exactly one box; they need not have zero measure in S . Summing the bounds with the finitely many absolute chain multiplicities proves the chain version. $\square$

6 Finite exceptional values and fiber periods

We prove the finite-exceptional-set statement by a direct scalar-gradient argument. No Newton-polytope hypothesis is needed in this section. Throughout, $d \geq 1$ and $f \in R_d$ is nonconstant. The two points to keep distinct are the proper geometry used for complete transport and the later semialgebraic representatives used for integration.

6.1 Closed embedding and the restricted differential

Put $T = (\mathbb{C}^\times)^d$ and define

$$\Phi : T \longrightarrow \mathbb{C}^{2d}, \quad \Phi(z) = (z_1, \dots, z_d, z_1^{-1}, \dots, z_d^{-1}). \quad (6.1)$$

Its image is the closed smooth affine variety

$$X = \{(z, w) : z_i w_i = 1 \ (1 \leq i \leq d)\}.$$

The defining differentials are independent, since their nonzero coordinate pairs are disjoint. Replacing negative powers by powers of w_i represents f by the restriction of a polynomial to X ; write $F : X \rightarrow \mathbb{C}$ for this restriction, so $F \circ \Phi = f$.

Use the Euclidean norm on $X \subset \mathbb{C}^{2d}$ and its induced Hermitian metric. In torus coordinates set

$$r(z) = \|\Phi(z)\| = \left(\sum_{i=1}^d (|z_i|^2 + |z_i|^{-2}) \right)^{1/2}. \quad (6.2)$$

For every R , the set $\{z : r(z) \leq R\}$ is compact in T . Indeed it is identified with a closed bounded subset of X ; equivalently all $|z_i|$ are bounded above and bounded away from zero. Thus an escaping sequence in T has $r(z_n) \rightarrow \infty$ precisely when it leaves every compact subset. The ordinary radius $\|z\|_{\mathbb{C}^d}$ would not have this property.

For $x \in X$ define the *restricted* differential norm

$$\lambda(x) = \|dF_x : T_x X \rightarrow \mathbb{C}\|. \quad (6.3)$$

This is not the norm of the differential of a chosen polynomial extension on all of $\mathbb{C}^{2d}$. Write $f_i = \partial f / \partial z_i$ and $h_i(z) = 1 + |z_i|^{-4}$. Direct differentiation of (6.1) and the weighted Cauchy–Schwarz inequality give

$$\|D\Phi_z(v)\|^2 = \sum_i h_i(z) |v_i|^2, \quad (6.4)$$

$$\lambda(\Phi(z))^2 = \sum_i \frac{|f_i(z)|^2}{h_i(z)}. \quad (6.5)$$

In particular, λ is continuous and semialgebraic. Because dF has one complex-linear output, it is either zero or surjective as a real map to $\mathbb{R}^2$. Moreover its operator norm controls both variation and the optimal right-inverse norm. Explicitly, whenever $\lambda(\Phi(z)) > 0$, for every $a \in \mathbb{C}$ the vector

$$V_a(z)_i = \frac{a \overline{f_i(z)}}{h_i(z) \lambda(\Phi(z))^2} \quad (6.6)$$

satisfies

$$df_z(V_a(z)) = a, \quad \|D\Phi_z(V_a(z))\| = \frac{|a|}{\lambda(\Phi(z))}. \quad (6.7)$$

These identities follow by substitution in (6.5); Cauchy–Schwarz also shows that no lift of a has smaller norm. For a general map with several outputs this equality between operator norm and least-surjectivity norm would fail. The scalar formula avoids any need for a matrix pseudoinverse.

6.2 Finitely many bad values

Define

$$K_0 = F(\{x \in X : \lambda(x) = 0\}), \quad (6.8)$$

$$K_\infty = \left\{ c \in \mathbb{C} : \begin{array}{l} \exists x_n \in X, \quad \|x_n\| \rightarrow \infty, \quad F(x_n) \rightarrow c, \\ \|x_n\| \lambda(x_n) \rightarrow 0 \end{array} \right\}. \quad (6.9)$$

Here K_0 is the set of ordinary critical values. Every occurrence of escape and of the differential norm in (6.9) uses (6.1)–(6.3).

Lemma 6.1 (Direct scalar finiteness). *Both K_0 and K_∞ are finite.*

Proof. The critical locus is semialgebraic. By Standard input 3.1, it has a finite decomposition into connected smooth pieces, after splitting the Nash pieces into their finitely many connected components. The restriction of F has zero differential on each such piece and is therefore constant there, by integration along piecewise smooth paths. Thus its image K_0 is finite.

For $R > 1$ and $\varepsilon > 0$, consider the compact semialgebraic family

$$A_{R,\varepsilon} = \{u \in \mathbb{S}^{4d-1} : Ru \in X, R\lambda(Ru) \leq \varepsilon\}. \quad (6.10)$$

The family is compact fiberwise because X is closed and λ is continuous. It is one fixed semialgebraic family by (6.5). Lemma 3.7 supplies $N \geq 1$ and $L \geq 1$ independent of both R and ε . If C is a connected component and $u, v \in C$, choose a connecting path of length at most L . On its arclength-parametrized arcs the scaled path $R\gamma$ lies in X , so its derivative is tangent to X almost everywhere and

$$|(F(R\gamma))'| \leq R\lambda(R\gamma) \|\gamma'\| \leq \varepsilon \|\gamma'\|.$$

Ordinary integration on the rectifiable arcs yields

$$\text{diam } F(RC) \leq L\varepsilon. \quad (6.11)$$

Suppose that K_∞ contains distinct $c_0, \dots, c_N$. Put $\eta = \min_{i \neq j} |c_i - c_j| > 0$ and choose $\varepsilon > 0$ with $L\varepsilon < \eta/4$. For each j let

$$E_j = \left\{ R > 1 : \begin{array}{l} \exists x \in X, \quad \|x\| = R, \quad R\lambda(x) \leq \varepsilon, \\ |F(x) - c_j| < \eta/8 \end{array} \right\}. \quad (6.12)$$

It is semialgebraic, by projection, and unbounded by the definition of K_∞ . Hence E_j contains a whole tail (R_j, ∞) . At a single radius $R > \max_j R_j$, choose the $N+1$ corresponding points x_j . The normalized points x_j/R lie in $A_{R,\varepsilon}$, while

$$|F(x_i) - F(x_j)| > \eta - 2\eta/8 = 3\eta/4 \quad (i \neq j).$$

Two normalized points must lie in the same one of at most N components, contradicting (6.11). Consequently $|K_\infty| \leq N$.

This proof uses semialgebraicity of the radius sets E_j , not a prior finiteness or algebraicity theorem for K_∞ . It does not require curve selection at infinity or Puiseux expansions. $\square$

We also write $K_0(F) = K_0$ and $K_\infty(F) = K_\infty$ when making the function explicit.

Proposition 6.2 (Finiteness of generalized critical values). *For nonconstant $f \in R_d$, the set*

$$B := K_0(F) \cup K_\infty(F) \subset \mathbb{C}$$

is finite.

Proof. This is the union of the two finite sets in Lemma 6.1. $\square$

Put $U = \mathbb{C} \setminus B$.

Lemma 6.3 (Uniform differential lower bound). *For every compact $Q \subset U$ there is $c_Q > 0$ such that*

$$(1 + \|x\|)\lambda(x) \geq c_Q \quad (x \in X, F(x) \in Q). \quad (6.13)$$

Proof. An empty inverse image is harmless. Otherwise, if no positive lower bound exists, choose x_n with $F(x_n) \in Q$ and $(1 + \|x_n\|)\lambda(x_n) \rightarrow 0$. Compactness gives a subsequence with $F(x_n) \rightarrow c \in Q$. If this subsequence has a bounded subsequence, closedness of X gives a further convergent subsequence with limit $x \in X$, $F(x) = c$, and $\lambda(x) = 0$. Then $c \in K_0$. If it has no bounded subsequence, its norms tend to infinity. The same inequalities then give $c \in K_\infty$. Both alternatives contradict $Q \cap B = \emptyset$. $\square$

6.3 Complete gradient transport

We use ordinary finite-dimensional real ODE theory on an open subset of $\mathbb{C}^d \simeq \mathbb{R}^{2d}$: a smooth vector field has unique maximal solutions, solutions depend smoothly on initial data and finite-dimensional parameters on their domain of existence, and a solution confined to a compact subset of the vector-field domain cannot have a finite terminal time. The same facts hold for time-dependent fields continuous in time and smooth in the spatial variables, with their spatial derivatives continuous; the relevant local Lipschitz bounds are uniform on compact time-space sets. These are the local existence, dependence, and continuation results of [15, Sections 2.2 and 2.4–2.6]. For the local trivializations below, dependence on finite-dimensional parameters is used only for the explicit smooth family of autonomous fields V_a . The growth estimate needed below is proved directly.

Lemma 6.4 (Transport along a segment). *Let $\sigma(t) = s_0 + ta$, $0 \leq t \leq 1$, be a segment contained in U . For every $z_0 \in f^{-1}(s_0)$, the real ODE $\dot{z} = V_a(z)$, $z(0) = z_0$, is defined throughout $[0, 1]$ and satisfies $f(z(t)) = \sigma(t)$. Its time-one map is a smooth diffeomorphism $f^{-1}(s_0) \rightarrow f^{-1}(s_0 + a)$. Transport of a compact cycle stays in a compact subset of T throughout this segment.*

Proof. The field (6.6) is real smooth on $\mathcal{O} = \{z \in T : \lambda(\Phi(z)) > 0\}$. The first identity in (6.7) shows that $f(z(t)) = s_0 + ta$ as long as the solution exists. Apply Lemma 6.3 with the compact set $Q = \sigma([0, 1])$. Along the existing part of the solution,

$$\left\| \frac{d}{dt} \Phi(z(t)) \right\| \leq \frac{|a|}{c_Q} (1 + r(z(t))). \quad (6.14)$$

Since $r(z) \geq \sqrt{2d} > 0$, its derivative exists along the smooth solution and is at most the norm on the left. Set $C = |a|/c_Q$. Differentiating $e^{-Ct}(1 + r(z(t)))$ gives a nonpositive quantity; hence

$$1 + r(z(t)) \leq (1 + r(z_0))e^{Ct} \quad (0 \leq t \leq 1) \quad (6.15)$$

whenever the solution exists.

Choose $R_* > (1 + r(z_0))e^C$. The trajectory, before any putative terminal time at most one, lies in

$$\mathcal{K} = \{z \in T : r(z) \leq R_*, f(z) \in Q\}.$$

This set is compact, and (6.13) gives $\lambda(\Phi(z)) \geq c_Q/(1 + R_*) > 0$ on it. Thus $\mathcal{K}$ is a compact subset of $\mathcal{O}$, not merely a bounded subset of T . ODE continuation excludes such a terminal time. In particular the bound controls both $|z_i|$ and $|z_i|^{-1}$: neither infinity nor a deleted coordinate hyperplane can be reached in finite time.

The reversed segment uses $V_{-a} = -V_a$; uniqueness gives the inverse of the time-one map. Smooth dependence gives smoothness of both maps. For initial points in a compact cycle, $r(z_0)$ has a common upper bound. The same argument is therefore uniform on that cycle, and its sweep over $[0, 1]$ is compact by continuous dependence. $\square$

For $s \in \mathbb{C}$, write $X_s = f^{-1}(s)$.

Proposition 6.5 (Complete ODE transport). *Let B be the finite set in Proposition 6.2 and put $U = \mathbb{C} \setminus B$. Every fiber over U is smooth. Moreover, for every piecewise C^1 path $\alpha : [0, 1] \rightarrow U$, the minimal-norm horizontal ODE gives a complete transport diffeomorphism*

$$X_{\alpha(0)} \simeq X_{\alpha(1)}.$$

In particular

$$f : f^{-1}(U) \longrightarrow U$$

is locally smoothly trivial, and its fiber homology groups form a local system on U .

Proof. Values outside K_0 have smooth fibers. First consider a C^1 piece $\alpha : [u, v] \rightarrow U$; here the derivative is continuous up to the endpoints. With $Q = \alpha([u, v])$, solve the real ODE on $\mathcal{O} = \{z \in T : \lambda(\Phi(z)) > 0\}$ given by

$$\dot{z}_i(t) = V_{\alpha'(t)}(z(t))_i = \frac{\alpha'(t) \overline{f_i(z(t))}}{h_i(z(t)) \lambda(\Phi(z(t)))^2}, \quad f(z(u)) = \alpha(u). \quad (6.16)$$

The coefficients are continuous in time and smooth in space, so local existence and uniqueness apply. By (6.7), $f(z(t)) = \alpha(t)$ while the solution exists. Put $C = \sup_{[u, v]} |\alpha'|/c_Q$. Then

$$\left\| \frac{d}{dt} \Phi(z(t)) \right\| \leq C(1 + r(z(t))), \quad 1 + r(z(t)) \leq (1 + r(z(u)))e^{C(t-u)}. \quad (6.17)$$

The second estimate follows by the same integrating-factor calculation as (6.15). As in Lemma 6.4, the trajectory stays in a compact set $\{r \leq R_*, f \in Q\}$ on which $\lambda \geq c_Q/(1 + R_*) > 0$. This is a compact subset of the ODE domain; continuation therefore covers the whole piece. The reversed path gives the inverse transport, and smooth dependence on the initial point gives a diffeomorphism of fibers. A piecewise C^1 path has a finite partition into such pieces; composing their transports proves the path assertion. Each lift is the minimal-norm lift by (6.7) and the Cauchy–Schwarz argument following it. For a polygonal path all the pieces use the autonomous fields of Lemma 6.4.

For local triviality, fix $b \in U$ and an open disk D centered at b with $\overline{D} \subset U$. For $s \in D$ let $\Theta(s, z)$ be the time-one solution of V_{s-b} starting at $z \in X_b$. Lemma 6.4 applies to every such segment, using the single compact base set $\overline{D}$ for the differential lower bound. Smooth dependence on s and the initial point makes

$$\Theta : D \times X_b \longrightarrow f^{-1}(D)$$

smooth. Its inverse sends y to $f(y)$ together with the time-one solution of $V_{b-f(y)}$ starting at y , and is smooth for the same reason. The reverse-flow identity proves that the maps are inverse; moreover $f(\Theta(s, z)) = s$. Empty fibers cause no difficulty, since reverse transport shows that a disk with empty reference fiber has empty preimage.

The induced fiber homology groups form a local system. No canonical or globally monodromy-invariant cycle is claimed. Uniformity on any fixed compact cycle follows from the radius bound and continuous dependence; no bound uniform over an entire noncompact fiber is required. $\square$

Remark 6.6 (Behavior at infinity is essential). Newton interiority does not let one replace B by K_0 . For $c \in \mathbb{C}$, consider

$$f_c(x, y) = c + x + \frac{(y-1)^2}{xy}.$$

Its Newton polytope has vertices $(1, 0), (-1, 1), (-1, -1)$, and the origin is their convex combination with positive coefficients $1/2, 1/4, 1/4$. Along $(x, y) = (t, 1)$, $t \downarrow 0$, one has

$$f_c(t, 1) = c + t, \quad (f_c)_x(t, 1) = 1, \quad (f_c)_y(t, 1) = 0,$$

and, in the metric (6.4),

$$\lambda(\Phi(t, 1)) = \frac{t^2}{\sqrt{1+t^4}}, \quad r(t, 1) = t + t^{-1}.$$

Thus $r(t, 1)\lambda(\Phi(t, 1)) \rightarrow 0$ and $c \in K_\infty$. The ordinary critical points have $y = -1$ and $x = \pm 2i$, with critical values $c \pm 4i$. Hence c is an ordinary regular value despite this asymptotic degeneration.

Remark 6.7 (Scope and attribution of the transport proof). The normalized-gradient mechanism is a scalar specialization of the asymptotic-critical-value approach to nonproper fibrations; see [12, Section 3] and [13]. The proof here establishes the needed finiteness by the common-radius argument and establishes completeness by (6.15), rather than invoking a generalized Bertini–Sard or fibration theorem. The tangency-value alternative in [14, Theorem 3.1] is related background, not an input to this proof. Appendix A retains the independent projective-graph/Whitney–Thom argument.

The real-smooth field V_a is semialgebraic, but its flow need not be: even $\dot{u} = u$ has an exponential flow. Transport provides the homology class and local trivializations, not the semialgebraic chain used for the endpoint estimate. Those representatives will be chosen separately in Section 7.

6.4 Relative forms and holomorphic periods

At a regular value s , put $X_s = f^{-1}(s)$. There is a canonical holomorphic $(d-1)$ -form η_s on X_s , denoted ω_d/df : locally choose a $(d-1)$ -form η with $df \wedge \eta = \omega_d$ and restrict it to the fiber. Differences between two such choices restrict to zero. In local holomorphic coordinates $(f, z'_2, \dots, z'_d)$, this is the ordinary relative holomorphic top form. In particular $d\eta_s = 0$ on the fiber.

Lemma 6.8 (Holomorphy of the transported period). *Let $[\gamma_s] \in H_{d-1}(X_s; \mathbb{Z})$ be a locally constant homology class over a simply connected open subset of U from Proposition 6.5. Then*

$$J(s) = \int_{\gamma_s} \eta_s$$

is well defined and holomorphic. Consequently its germ can be continued along any path in U .

Proof. Closedness of η_s and the ordinary de Rham pairing show that the integral depends only on the homology class. Proposition 6.5 provides smooth local trivializations by complete horizontal ODE transport. Choose a compact smooth singular cycle in one fiber and transport it in a small disc; all transported cycles remain in the image of one compact cycle under a smooth map from a compact parameter set.

For a small oriented triangle Δ in the disc, use such a local trivialization to sweep the cycle over Δ . This is a $(d+1)$ -chain Γ_Δ , whose boundary is the sweep over $\partial\Delta$. On each edge, $\omega_d = df \wedge \eta$ and f is the base coordinate, so ordinary parametrized integration gives

$$\int_{\partial\Delta} J(s) ds = \int_{\partial\Gamma_\Delta} \omega_d = \int_{\Gamma_\Delta} d\omega_d = 0.$$

The holomorphic top form ω_d is closed. Continuity of J follows from the same compact smooth family, and Morera’s theorem proves holomorphy. Repeating this along overlapping discs gives continuation along every path in U . No global monodromy invariance of the transported class is required. $\square$

7 Semialgebraic sweeps and the endpoint primitive

The flows of Section 6 select the transported homology class. They are not used as semialgebraic parametrizations. We now use Hardt triviality again, independently of its earlier use in Lemma 3.7, to obtain the required semialgebraic sweep. Continue to assume $0 \in \text{int Newt}(f)$ and $d \geq 1$. Choose $b > 0$ sufficiently small that $(0, b] \subset U$. Applying Hardt triviality to the restriction of f over the real interval, and decreasing b once more if necessary, gives a semialgebraic homeomorphism

$$\Psi : (0, b] \times X_b \longrightarrow f^{-1}((0, b]), \quad f(\Psi(s, z)) = s, \quad \Psi(b, z) = z. \quad (7.1)$$

The last normalization is obtained by composing the fiber coordinate with the inverse of its value at b .

Fix a compact semialgebraic cycle γ_b representing the prescribed transported class at b , and put $\gamma_s = \Psi(s, \gamma_b)$. These classes coincide with those used for analytic continuation. Indeed, both are locally constant in the same fiber-homology local system and agree at b on the connected interval. Thus their periods are the function J of Lemma 6.8.

Triangulate the finite cycle γ_b and sweep its simplices using (7.1). Denote the resulting finite semialgebraic d -chain by S , and its restriction over $[a, c] \subset (0, b]$ by $S_{[a, c]}$. Orient the base first, so

$$\partial S_{[a, c]} = \gamma_c - \gamma_a. \quad (7.2)$$

Shared simplex faces cancel with their cycle multiplicities. Each truncated sweep is compact: it is the continuous image of a finite compact cycle times $[a, c]$. Its support lies in the regular locus of f . The full sweep need not have compact closure in the torus.

Proposition 7.1 (Quantitative endpoint primitive). *There are $C > 0$, $\delta > 0$, and a function $I : (0, b] \rightarrow \mathbb{C}$, continuous on $(0, b]$ and C^1 on $(0, b)$, such that*

$$I'(s) = J(s), \quad (7.3)$$

and I extends continuously to 0 with

$$|I(t) - I(0)| \leq Ct^\delta \quad (0 < t \leq b). \quad (7.4)$$

In particular, for $0 < a < c \leq b$,

$$\int_a^c J(s) ds = \int_{S_{[a, c]}} \omega_d, \quad (7.5)$$

and, after an enlargement of C if needed,

$$\left| \int_a^{2a} J(s) ds \right| \leq C(2a)^\delta \quad (0 < 2a \leq b). \quad (7.6)$$

Proof. By Theorem 5.1, with $\delta = 1/D$ as there,

$$\int_{|S| \cap \{|f| \leq r\}} |\omega_d| \leq Cr^\delta \quad (r > 0). \quad (7.7)$$

This applies separately to the finitely many swept simplices and then with their absolute multiplicities. In particular the integral over the whole sweep is absolutely convergent, since $|f| \leq b$ there. Define

$$I(t) = - \int_{S_{[t, b]}} \omega_d, \quad I(0) = - \int_S \omega_d.$$

The possible endpoint slices have dimension at most $d - 1$ and contribute zero to these d -integrals. Since $f = s$ on the sweep, (7.7) proves (7.4). It remains to prove (7.3); the fundamental theorem of calculus will then give (7.5).

For this local Stokes calculation only, no control of escape is needed. The following lift in the original coordinates is distinct from the proper-metric transport field (6.6). On the regular locus of f , define the smooth complex vector field and smooth $(d - 1)$ -form

$$V = \frac{\sum_{j=1}^d \overline{\partial_j f} \partial / \partial x_j}{\sum_{j=1}^d |\partial_j f|^2}, \quad \eta = \iota_V \omega_d. \quad (7.8)$$

Then $df(V) = 1$ and $df \wedge \eta = \omega_d$; the latter follows by contracting the identically zero $(d+1)$ -form $df \wedge \omega_d$. The restriction of η to X_s is η_s .

For $0 < u < v \leq b$, the identity

$$\omega_d = d((f - u)\eta) - (f - u)d\eta$$

and Stokes on the compact semialgebraic sweep give

$$I(v) - I(u) = (v - u)J(v) - \int_{S_{[u,v]}} (f - u)d\eta. \quad (7.9)$$

The lower boundary term is zero, and the upper one is $(v - u) \int_{\gamma_v} \eta_v$. This is the local differentiation identity in [3, Lemma 9].

Fix $s_0 \in (0, b)$. All sweeps over a sufficiently small closed neighborhood of s_0 lie in one compact subset of the regular locus. There $d\eta$ has bounded norm. Since f takes values in $[u, v]$ on $S_{[u,v]}$, the error in the difference quotient is bounded by

$$\left| \frac{I(v) - I(u)}{v - u} - J(v) \right| \leq C_{s_0} \text{mass}_d(S_{[u,v]}). \quad (7.10)$$

Here mass means the sum of the Euclidean d -volumes of the fixed swept pieces with their absolute multiplicities. It is finite on each fixed compact truncation by Corollary 3.6. As $v \downarrow u = s_0$, the sets decrease to a $(d-1)$ -dimensional slice, so this mass tends to zero by continuity from above of a finite measure. The same argument applies from the other side. Continuity of J , proved independently in Lemma 6.8, yields $I'(s_0) = J(s_0)$. Thus I is C^1 on $(0, b)$. The same compact-truncation argument at the upper endpoint gives $I(t) \rightarrow I(b) = 0$ as $t \uparrow b$.

Finally, (7.5) and (7.7), applied to $S_{[a, 2a]} \subset \{|f| \leq 2a\}$, prove (7.6). No limiting Stokes theorem at $s = 0$ has been used: Stokes was applied only on compact truncations away from zero. $\square$

Corollary 7.2 (Exclusion of the forced simple pole). *The period in Proposition 7.1 cannot satisfy $J(s) = A/s$ on $(0, b]$ for any constant $A \neq 0$.*

Proof. For $0 < 2a \leq b$,

$$\int_a^{2a} \frac{A}{s} ds = A \log 2.$$

Its modulus is the positive constant $|A| \log 2$, whereas (7.6) tends to zero as $a \downarrow 0$. $\square$

8 The local residue identity and minimal nonvanishing

By Lemmas 2.1 and 2.3, the only case still requiring proof is $d \geq 1$, $0 \in \text{int Newt}(f)$, and

$$f(y) = y^{-m}u(y), \quad m_i \geq 1, \quad u(0) \neq 0.$$

All constant terms are computed in these transformed variables, where they equal the original constant terms. Set

$$\omega = \frac{dy_1}{y_1} \wedge \cdots \wedge \frac{dy_d}{y_d}.$$

Lemma 8.1 (Residue cycles near a vertex). *For all sufficiently large $|s|$, there is a compact semialgebraic $(d-1)$ -cycle $\gamma_s \subset f^{-1}(s)$, varying locally smoothly, such that*

$$R(s) := \sum_{n \geq 0} \frac{\text{CT}(f^n)}{s^{n+1}} = -(2\pi i)^{1-d} \int_{\gamma_s} \frac{\omega}{df}. \quad (8.1)$$

The series converges for all sufficiently large $|s|$.

Proof. Choose $\varepsilon > 0$ so small that u has no zero on the closed polydisc $|y_i| \leq \varepsilon$ and

$$|y_1 \partial_1 u| < m_1 |u| \quad \text{on that polydisc.} \quad (8.2)$$

Let $K_\varepsilon = \{|y_i| = \varepsilon \ (1 \leq i \leq d)\}$ with its ordered product-circle orientation. If $|s| > \max_{K_\varepsilon} |f|$, the geometric series is uniformly convergent on K_ε and coefficient extraction gives

$$R(s) = \frac{1}{(2\pi i)^d} \int_{K_\varepsilon} \frac{\omega}{s - f}. \quad (8.3)$$

Indeed the integral of each Laurent monomial is zero unless its exponent is zero, in which case it equals $(2\pi i)^d$.

For $y' = (y_2, \dots, y_d)$ on the product circle $K' = \{|y_j| = \varepsilon \ (j > 1)\}$, solve

$$s y_1^{m_1} \prod_{j>1} y_j^{m_j} - u(y_1, y') = 0. \quad (8.4)$$

For $|s|$ uniformly large, Rouché's theorem on $|y_1| = \varepsilon$ shows that there are exactly m_1 roots in the disc, counted with multiplicity. None is zero because $u(0, y') \neq 0$. At a root, the y_1 -derivative of the left side of (8.4) is $(m_1 u - y_1 \partial_1 u)/y_1$, which is nonzero by (8.2). Thus all roots are simple.

Let γ_s be the union of these roots over K' . It is a compact unbranched m_1 -sheeted cover of K' , oriented by the local projections onto the ordered base circles. Compactness follows uniformly from the absence of roots on either $y_1 = 0$ or $|y_1| = \varepsilon$ and compactness of K' . The defining equations and inequalities are semialgebraic. The cover has no boundary and so defines a cycle. For $d = 1$, interpret K' as a point and γ_s as the positively weighted finite set of roots.

As a one-variable form in y_1 ,

$$\frac{dy_1}{y_1(s - f)} = \frac{y_1^{m_1-1} (\prod_{j>1} y_j^{m_j}) dy_1}{s y_1^{m_1} \prod_{j>1} y_j^{m_j} - u(y)}.$$

It is holomorphic at $y_1 = 0$. At a root ζ its residue is $-1/(\zeta \partial_1 f(\zeta, y'))$. On this part of the fiber,

$$\frac{\omega}{df} = \frac{1}{y_1 \partial_1 f} \frac{dy_2}{y_2} \wedge \dots \wedge \frac{dy_d}{y_d}.$$

Integrating first in y_1 and then over K' gives

$$\int_{K_\varepsilon} \frac{\omega}{s - f} = -2\pi i \int_{\gamma_s} \frac{\omega}{df}.$$

Together with (8.3), this proves (8.1), including its sign and normalization. The implicit function theorem at the simple roots gives local smooth variation of the cycle and the associated locally constant homology class. $\square$

Proof of Theorem 1.2. If the minimal-face reduction has rank zero, f_Q is a nonzero constant and its first constant term is nonzero. Otherwise apply the reduction and unimodular change just described. Suppose for contradiction that $\text{CT}(f^n) = 0$ for every $n \geq 1$. Lemma 8.1 then gives

$$R(s) = \frac{1}{s}, \quad J(s) := \int_{\gamma_s} \frac{\omega}{df} = -\frac{(2\pi i)^{d-1}}{s}$$

for all sufficiently large $|s|$.

Let B be the finite set in Proposition 6.5. Choose a large base point outside $B \cup \{0\}$ and continue the homology class of the residue cycle along a path in $\mathbb{C} \setminus (B \cup \{0\})$ to a sufficiently small positive b . Such a path exists because the complement of finitely many points in $\mathbb{C}$ is path connected. Lemma 6.8 continues the scalar period. Since the rational function $-(2\pi i)^{d-1}/s$ is holomorphic along the same path, uniqueness of analytic continuation gives

$$J(s) = -\frac{(2\pi i)^{d-1}}{s} \quad (0 < s \leq b).$$

Here J denotes the period of the continued homology class, not the integral over the original fixed product torus after leaving its initial convergence region. The cycle may have monodromy; the displayed scalar identity is what is continued.

Proposition 7.1 applies to this class and its semialgebraic representatives on $(0, b]$. Corollary 7.2 contradicts the displayed identity, since $(2\pi i)^{d-1} \neq 0$. The assumed universal vanishing is therefore impossible. $\square$

9 Infinite nonvanishing and the torus Mathieu property

Proof of Theorem 1.1. Fix $M \geq 0$ and set $q = M + 1$. By Lemma 2.2, $0 \in \text{Newt}(f^q)$. Theorem 1.2 gives $k \geq 1$ with

$$0 \neq \text{CT}((f^q)^k) = \text{CT}(f^{qk}).$$

Since $qk \geq q > M$, this proves the assertion. In particular, no separate argument concerning higher-order poles is required. $\square$

Corollary 9.1 (Newton-polytope classification). *For $0 \neq f \in R_d$,*

$$\text{CT}(f^n) = 0 \text{ for every } n \geq 1 \iff 0 \notin \text{Newt}(f).$$

Proof. One direction is the contrapositive of Theorem 1.2. For the other, strict separation of a point and a compact convex set gives a linear functional ℓ and $\delta > 0$ with $\ell(\alpha) \geq \delta$ for every $\alpha \in \text{supp}(f)$. Every exponent in f^n is a sum of n such exponents, and has ℓ -value at least $n\delta > 0$. The zero exponent cannot occur. $\square$

Corollary 9.2 (Laurent-polynomial Mathieu property). *Let $f, h \in R_d$. If $\text{CT}(f^n) = 0$ for every $n \geq 1$, then $\text{CT}(hf^n) = 0$ for all sufficiently large n .*

Proof. The assertion is immediate if $f = 0$ or $h = 0$. Otherwise use Corollary 9.1 and choose ℓ, δ as in its proof. Put $A_h = \min_{\beta \in \text{supp}(h)} \ell(\beta)$. Any exponent in hf^n has ℓ -value at least $A_h + n\delta$, which is positive for all sufficiently large n . Thus its constant coefficient is zero. $\square$

Corollary 9.3 (Compact tori). *Let $K = \mathbb{T}^d$ with normalized Haar measure. For finite Fourier sums F, H on K , if $\int_K F^n = 0$ for every $n \geq 1$, then $\int_K HF^n = 0$ for all sufficiently large n .*

Proof. Identify a character with $x^\alpha = e^{i\langle \alpha, \theta \rangle}$, $\alpha \in \mathbb{Z}^d$. Direct integration in each circle coordinate gives

$$\frac{1}{(2\pi)^d} \int_{[0, 2\pi]^d} e^{i\langle \alpha, \theta \rangle} d\theta = \begin{cases} 1, & \alpha = 0, \\ 0, & \alpha \neq 0. \end{cases}$$

Hence the Haar integral of any Laurent polynomial restricted to K is its constant coefficient. Apply Corollary 9.2. $\square$

Remark 9.4. Finite Fourier sums are exactly the continuous K -finite functions on a torus. One direction follows because each character spans a one-dimensional translation module. Conversely, on a finite-dimensional translation-invariant space, average an inner product over K and simultaneously diagonalize the commuting unitary translations. An eigenfunction is a scalar multiple of the corresponding character, by evaluation at the identity. The continuous characters of $\mathbb{T}^d$ are precisely those indexed by $\mathbb{Z}^d$. This is the usual torus interpretation of [3]; no assertion about nonabelian groups is needed in this paper.

10 Provenance and mathematical scope

The theorem proved is the arbitrary-rank nonvanishing theorem of Duistermaat and van der Kallen. The residue construction in Lemma 8.1 is their local construction, and the calculation (7.9) is their primitive-differentiation identity. The contribution proposed for independent examination is the uniform middle-dimensional sublevel estimate and its dyadic use in Theorem 5.1. This replaces the argument making the Haar form regular over finite fibers of a smooth compactification.

The semialgebraic integration estimates in [6] are antecedents, not conclusions attributed to the present paper. The coefficient-derivative induction in Section 4 is also naturally compared with geometric sublevel-set estimates; see, for example, [5] and the references there. The specific estimate here concerns the modulus of a restricted holomorphic top form and is uniform in a fixed semialgebraic family. These precise hypotheses, rather than merely similar terminology, must be used in any priority comparison.

Question 9.1 of [9] motivates the search for a simpler nonvanishing proof. The main route retains substantial, explicitly identified semialgebraic and integration results, together with ordinary finite-dimensional ODE theory. Whitney stratification and Thom isotopy are confined to the independent appendix proof and are not dependencies of the main route. The argument is therefore described as resolution-free rather than as elementary in every respect. The stronger singularity and growth conclusions of [3, Theorem 5] are outside the claims made here. A proof and generalization using Newton–Puiseux methods in one variable is discussed in [4]; the present theorem has no rank restriction.

The intended future formalization has one core endpoint, Theorem 1.2. Lemma 2.2 and the separation argument then give Theorem 1.1 and Corollary 9.2. The semialgebraic and integration inputs in Section 3 and the ODE facts identified in Section 6 remain genuine mathematical dependencies: a formal proof must establish them or invoke proved library theorems, not add them as unproved project axioms. In particular, the connecting-path lemma is derived from the existing uniform-volume bound; finiteness of K_∞ is then proved rather than supplied by an external theorem. Formalization of the independent appendix proof would additionally need Standard input A.1; that input is not required to formalize the main scalar-gradient route.

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A Whitney–Thom alternative for finite bifurcation and cycle transport

This appendix records an independent proof of the finite-exceptional-set and topological transport conclusion needed for the period argument. It is not used in the main proof. The exceptional set below need not equal $K_0 \cup K_\infty$. This route does not assert the quantitative gradient bounds or the specific transport diffeomorphisms of Proposition 6.5.

Standard input A.1 (Complex algebraic stratification and first isotopy). A complex algebraic variety, together with finitely many specified algebraic subsets, admits a finite complex algebraic Whitney stratification compatible with those subsets. If a map from a closed Whitney-stratified subset of a smooth manifold to a smooth base is proper and restricts to a submersion on every stratum, it is locally topologically trivial by homeomorphisms preserving the strata.

These are the algebraic Whitney-stratification theorem and Thom’s first isotopy lemma; see [8, 10]. For a direct treatment of nonproper polynomial maps, see [2], especially Theorems 3.1 and 5.1 of its arXiv version.

Proposition A.2 (Whitney–Thom finite exceptional set). *For a nonconstant $f \in R_d$, there is a finite set $B_W \subset \mathbb{C}$ such that all fibers over $U_W = \mathbb{C} \setminus B_W$ are smooth and*

$$f : f^{-1}(U_W) \longrightarrow U_W$$

is locally topologically trivial.

Proof. Let $\bar{\Gamma}$ be the closure of the graph of f in $(\mathbb{P}^1)^d \times \mathbb{P}^1$, let $p : \bar{\Gamma} \rightarrow \mathbb{P}^1$ be projection, and let D_∞ be the complement of the open graph identified with $(\mathbb{C}^\times)^d$. The projection p is proper. Choose a finite complex algebraic Whitney stratification compatible with D_∞ .

For a stratum W , the critical locus of $p|_W$ is algebraic in W . At such a point its complex differential has rank zero, because the base has complex dimension one. On the smooth locus of any irreducible component of this critical locus, the restriction of p consequently has zero differential and is locally constant. It is therefore constant on that component. Equivalently, a rational function with zero differential on an irreducible complex variety is constant. There are only finitely many components and strata, so their critical images form a finite subset of $\mathbb{P}^1$.

Remove that finite subset and the value ∞ . On the remaining base, p is a proper stratified submersion, so Standard input A.1 gives local trivializations preserving all strata. They preserve D_∞ and its complement. Restricting to the latter gives the claimed local trivializations of f . At a point of this open graph, submersivity on its stratum implies submersivity on the ambient smooth torus; hence every remaining fiber is smooth. Empty fibers, if present, cause no difficulty in this statement. In the application a nonempty fiber is supplied by the explicit residue construction. $\square$

Remark A.3. The adjective *complex algebraic* matters in this appendix. Real semialgebraic strata can map to curves of real critical values. Such a real exceptional set need not leave a connected complement through which the required analytic germ can be transported. The projective graph may be singular; no resolution theorem is hidden in this argument.

Remark A.4. Proposition A.2 can replace Propositions 6.2–6.5 in the nonvanishing proof. For the period itself, one then uses the local compact-cycle argument: on a sufficiently small disc of regular values, lift the two real coordinate vector fields only near the compact cycle support, transport the finite cycle smoothly, and apply the triangle-sweep/Morera argument of Lemma 6.8. Thus the appendix route supplies the same analytic continuation without any appeal to the ODE completeness argument of Section 6.

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